

The Infrared Structure of the Ω^0 and Ω^2 Bosonic Sectors of the Hodge Complex: Finite-Volume Scaling of the Boson Gap and Completion of the Hodge IR Picture

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Abstract

This paper completes the infrared analysis of the Hodge complex on the 8×8 torus begun in the companion paper on the Ω^1 fermionic sector. The Hodge complex consists of three graded pieces: the scalar Ω^0 , the 1-form Ω^1 (fermionic), and the pseudo-scalar Ω^2 . We characterize the bosonic sectors Ω^0 and Ω^2 directly. *First*, the free bosonic spectrum is computed exactly: Ω^0 and Ω^2 have identical spectra by Hodge duality, with a single massless mode in each and a free mass gap of $\sqrt{4 - \sqrt{2}} \approx 0.765$ in lattice units for the lightest massive excitation. *Second*, the natural D_4 -invariant quartic interactions on the bosonic sectors are the standard ϕ^4 and χ^4 interactions, which are site-local and preserve the D_4 symmetry. We analyze the question of cross-couplings between graded sectors: mixed quartic vertices such as $\phi^2\omega^2$ are permitted by the $D_4 \times \mathbb{Z}_2$ symmetry of the Hodge complex, but the natural interaction organizes into a Wess–Zumino superpotential for a single complex scalar $\Phi = \Omega^0 + i\Omega^2$. Under the superpotential structure, the same-sector quartics $|\Phi|^4$ dominate and the mixed cross-sector couplings to Ω^1 are generated only at higher order in the effective expansion. *Third*, we compute the one loop boson gap in the finite-volume limit $L \rightarrow \infty$ at fixed lattice spacing $a = 1$ and show that the leading L -dependence is generated by the IR modes $q \sim 2\pi/L$ and grows logarithmically:

$$m_{\text{eff}}^2(L) = 12 g \langle \phi^2 \rangle_{\text{free}}(L) = \frac{96 g}{\pi^3} \log L + O(1).$$

At the physical torus size $L = 8$, this evaluates to $m_{\text{eff}}^2 = 4.55 g$, matching the Hodge paper’s value and now derived rather than assumed. *Fourth*, we prove that the bosons decouple from IR observables on sufficiently large tori: their propagator at IR momenta $q \sim 2\pi/L$ is dominated by the mass term, and boson contributions to IR correlators are suppressed by $1/(L^2 \log L)$ relative to the leading free-field contribution. The deep IR of the Hodge complex is therefore controlled by the fermionic Ω^1 sector alone, which does not flow to SYK [16]. The FDLM/WIN processor operates on the mid-spectrum $\lambda = 4$ eigenspace, unaffected by the deep IR dynamics of either sector.

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1 Introduction

The Hodge complex on the 8×8 torus [11, 12] is the fundamental substrate of the WIN Paradigm. It consists of three graded pieces

$$\Omega^0 \oplus \Omega^1 \oplus \Omega^2,$$

with total dimension $64 + 128 + 64 = 256$, and carries the Hodge-Dirac operator $D = d + d^*$ satisfying $D^2 = \Delta$ exactly on each form degree. The even sectors Ω^0 and Ω^2 are bosonic; the odd sector Ω^1 is fermionic.

The companion paper [16] characterized the infrared structure of the fermionic Ω^1 sector. It established that the free Ω^1 spectrum has two massless modes (harmonic 1-forms) and a gap of $\sqrt{4 - \sqrt{2}} \approx 0.765$ in lattice units; that the natural D_4 -invariant quartic is site local; and that local interactions do not flow to SYK because momentum conservation at each vertex prevents melonic routing.

This paper fills the remaining gap. Its purpose is to complete the infrared picture of the Hodge complex by characterizing the bosonic sectors Ω^0 and Ω^2 the same way [16] characterized Ω^1 . Four results are established:

1. **Free bosonic spectrum** (§3). Ω^0 and Ω^2 have identical spectra by Hodge duality, with a single massless mode in each and the same free mass gap $\sqrt{4 - \sqrt{2}} \approx 0.765$.
2. **D_4 -invariant quartic and cross-couplings** (§4). The natural interaction is the site-local ϕ^4 (resp. χ^4) on each sector, plus cross-couplings between sectors. We prove that mixed quartics are permitted by $D_4 \times \mathbb{Z}_2$ but are suppressed relative to same-sector quartics in the natural superpotential structure of the Hodge complex.
3. **Finite-volume scaling of the boson gap** (§5). We compute the one-loop self-energy of the scalar and pseudo-scalar sectors and establish

$$m_{\text{eff}}^2(L) = 12 g \langle \phi^2 \rangle_{\text{free}}(L) \sim \frac{96 g}{\pi^3} \log L$$

at fixed lattice spacing $a = 1$. At $L = 8$, this evaluates to $m_{\text{eff}}^2 = 4.55 g$, matching the Hodge paper's value and now derived rather than assumed.

4. **Infrared suppression of the bosons** (§6). On sufficiently large tori, the boson propagator at IR momenta $q \sim 2\pi/L$ is dominated by the mass term, and boson contributions to IR observables are suppressed by $1/(L^2 \log L)$. The IR is controlled by the fermionic sector.

Together with [16], this paper completes the infrared analysis of the Hodge complex. Neither the bosonic nor the fermionic sectors flow to SYK. The FDLM/WIN processor operates on the mid-spectrum $\lambda = 4$ eigenspace, unaffected by the deep IR dynamics of either sector.

The paper is organized as follows. §2 sets up the field theoretic framework, including the lattice-spacing convention. §3 computes the free bosonic spectrum. §4 states the D_4 -invariant quartic interactions, discusses cross-couplings, and introduces the superpotential structure. §5 derives the boson gap and its finite-volume scaling, with explicit UV/IR language. §6 proves IR suppression. §7 discusses the consequences for the WIN architecture. §8 lists open questions. §9 concludes. Appendices give numerical verification of the free spectrum and the one-loop self-energy.

2 Field theoretic setup

We begin by stating the field theoretic setting precisely, including the lattice spacing convention used throughout.

2.1 The bosonic sectors

The Hodge complex on the spatial torus $\Lambda = \mathbb{Z}_L \times \mathbb{Z}_L$ has three graded pieces. For the physical case $L = 8$:

$$\Omega^0 = \mathbb{R}^{L^2}, \quad (\text{scalar functions}), \quad (1)$$

$$\Omega^1 = \mathbb{R}^{2L^2}, \quad (1\text{-forms; 2 components per site}), \quad (2)$$

$$\Omega^2 = \mathbb{R}^{L^2}, \quad (\text{pseudo-scalars; 1 component per site}). \quad (3)$$

The even sectors Ω^0 and Ω^2 carry bosonic statistics. The Hodge–Dirac operator $D = d + d^*$ acts on the full complex, with

$$D^2|_{\Omega^k} = \Delta \cdot \mathbb{K}_{\Omega^k}, \quad k = 0, 1, 2, \quad (4)$$

where Δ is the scalar 5-point Laplacian

$$(\Delta f)(x) = 4f(x) - \sum_{|e|=1} f(x+e). \quad (5)$$

2.2 Euclidean time and spacetime dimension

We attach a Euclidean time coordinate $\tau \in [0, \beta)$ and consider the path integral

$$Z = \int \mathcal{D}[\phi] e^{-S_E[\phi]}, \quad (6)$$

with Euclidean action

$$S_E[\phi] = \int_0^\beta d\tau \sum_{x \in \Lambda} \mathcal{L}(\phi(x, \tau), \partial_\tau \phi(x, \tau), \nabla \phi(x, \tau)). \quad (7)$$

The effective theory lives on a Euclidean spacetime of dimension

$$D_E = 1 + 2 = 3, \quad (8)$$

equivalently $(2+1)$ -dimensional in Lorentzian signature.

The Euclidean Lagrangian density for the scalar sector is

$$\mathcal{L} = \frac{1}{2}(\partial_\tau \phi)^2 + \frac{1}{2}|\nabla \phi|^2 + \frac{g}{4!}\phi^4, \quad (9)$$

and identically for the pseudo-scalar χ . In $D_E = 3$, the coupling has mass dimension $[g] = 1$, making the quartic interaction super-renormalizable.

2.3 Lattice-spacing convention

Remark 1 (Lattice-spacing convention). Throughout, unless otherwise stated, we work in *lattice units* with $a = 1$. All masses, gaps, and couplings are expressed in these units. The finite-volume limit studied in this paper is

$$\text{fixed } a = 1, \quad L \rightarrow \infty, \quad (10)$$

in which the UV cutoff is fixed at the Brillouin-zone boundary $q_{\max} \sim \pi$ while the IR cutoff $q_{\min} = 2\pi/L$ is lowered. This is the standard finite-volume limit of lattice field theory. We are *not* taking the continuum limit $a \rightarrow 0$ in this paper; the distinction is essential for the interpretation of the logarithmic growth of the boson gap established in §5.

Remark 2 (Continuum limit). A separate, physically distinct limit is the *continuum limit*, in which $a \rightarrow 0$ with the physical torus side $R = La$ held fixed. In this limit, additional UV divergences appear at the Brillouin-zone boundary and must be removed by standard mass and coupling renormalization. The super renormalizability of ϕ^4 in $D_E = 3$ guarantees that only finitely many counterterms are required. We do not analyze this limit in detail; the finite-volume scaling of the gap is the primary result of §5.

2.4 Bosonic field content

The dynamical field on Ω^0 is a real scalar $\phi(x, \tau)$ per site. The dynamical field on Ω^2 is a real pseudo-scalar $\chi(x, \tau)$ per site. Both are real bosonic fields with canonical kinetic terms.

Remark 3 (Hodge duality between Ω^0 and Ω^2). On a 2-dimensional torus, the Hodge star operator $\star$ maps 0-forms to 2-forms and vice versa: $\star : \Omega^0 \rightarrow \Omega^2$ is an isomorphism. Consequently Ω^0 and Ω^2 are equivalent as D_4 representations and as free field theories. The two sectors have identical free spectra, identical free gaps, and identical free structure. We treat them separately for clarity but note throughout that they are Hodge-dual.

Remark 4 (Complex scalar structure). The combination

$$\Phi(x, \tau) := \phi(x, \tau) + i\chi(x, \tau) \quad (11)$$

is a complex scalar field on the torus. Its real part is the Ω^0 scalar and its imaginary part is the Ω^2 pseudo-scalar. The free theory has a global $U(1)$ symmetry under $\Phi \rightarrow e^{i\theta}\Phi$, which acts as a simultaneous rotation of ϕ and χ . This structure plays a central role in the superpotential analysis of §4.3.

3 The free bosonic spectrum

We work on the spatial torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = L^2 = 64$ sites. The scalar 5-point Laplacian has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (12)$$

for $(n_1, n_2) \in \mathbb{Z}_8^2$. The same spectrum applies to Ω^0 and Ω^2 by Remark 3.

3.1 Spectrum

Table 1 lists the full scalar spectrum.

3.2 The single massless mode

Theorem 5 (Zero modes of the bosonic sectors). *Each of Ω^0 and Ω^2 has exactly one massless mode, the constant function*

$$\phi_0(x) = \text{const}, \quad \chi_0(x) = \text{const}. \quad (13)$$

The kernel of Δ on each sector has dimension 1, matching the zeroth Betti number $b_0(T^2) = 1$.

Proof. The kernel of the scalar 5-point Laplacian on the torus consists of functions f satisfying $f(x + e) = f(x)$ for all x and all unit vectors e . This forces f to be constant. The same argument applies to Ω^2 by Hodge duality. $\square$

λ	Multiplicity	$\sqrt{\lambda}$ (lattice mass)
0.0000	1	0.000
0.5858	4	0.765
1.1716	4	1.082
2.0000	4	1.414
2.5858	8	1.608
3.4142	4	1.848
4.0000	14	2.000
4.5858	4	2.141
5.4142	8	2.327
6.0000	4	2.449
6.8284	4	2.613
7.4142	4	2.723
8.0000	1	2.828
Total	64	

Table 1: Free spectrum of the scalar Laplacian on the 8×8 torus, with lattice spacing $a = 1$. The same spectrum applies to Ω^0 and (by Hodge duality) Ω^2 .

Remark 6. The massless mode of Ω^0 is the constant function, corresponding to the $N/4 = 60$ mode (the neutrino mode) in the FDLN/WIN particle multiplet. The massless mode of Ω^2 is its Hodge dual, the constant pseudo-scalar. Together they form the zero mode of the complex scalar Φ .

3.3 The lightest massive modes and the free gap

The lightest non-zero eigenvalue is

$$\lambda_{\min} = 4 - 2 \cos(\pi/4) - 2 \cos(0) = 4 - \sqrt{2} \approx 0.5858. \quad (14)$$

The corresponding free mass gap is

$$m_{\text{gap}}^{\text{free}} := \sqrt{\lambda_{\min}} = \sqrt{4 - \sqrt{2}} \approx 0.7654 \quad (15)$$

in lattice units.

The 4 modes at $\lambda = 0.5858$ in each bosonic sector are the lightest massive excitations.

4 Interactions and cross-couplings

The bosonic sectors are free unless we introduce interactions. In this section we first state the natural quartic interactions, then analyze the question of cross-couplings between sectors, and finally introduce the superpotential structure that organizes the interactions.

4.1 The same-sector quartic

Since each bosonic sector carries a single real scalar field per site, the unique D_4 -invariant quartic on each is the standard ϕ^4 interaction:

$$S_{\text{int}}^{\Omega^0}[\phi] = \int_0^\beta d\tau \sum_{x \in \Lambda} \frac{g}{4!} \phi(x, \tau)^4, \quad (16)$$

and identically for χ . The coupling has mass dimension 1 in $D_E = 3$.

Proposition 7 (Locality). *The interactions (16) are site-local: they are sums over sites of a polynomial in the field at that site, with no coupling between different spatial sites.*

4.2 Cross-couplings between sectors

We now address the question of mixed quartic vertices between the graded sectors. This question is essential because a mixed vertex $\phi^2\omega^2$ would, at one loop, generate a logarithmic mass correction for the fermions as well, modifying the fermionic gap from its fixed free value.

Proposition 8 (Allowed cross-couplings). *The following mixed quartic vertices are invariant under the $D_4 \times \mathbb{Z}_2$ symmetry of the Hodge complex:*

$$V_{\phi\omega}(x) = \phi(x)^2 [\omega_1(x)^2 + \omega_2(x)^2], \quad (17)$$

$$V_{\phi\chi}(x) = \phi(x)^2 \chi(x)^2, \quad (18)$$

$$V_{\chi\omega}(x) = \chi(x)^2 [\omega_1(x)^2 + \omega_2(x)^2]. \quad (19)$$

More generally, any product of D_4 singlets formed from the fields is permitted.

Proof. The $\mathbb{Z}_2$ grading assigns +1 to Ω^0, Ω^2 (bosons) and -1 to Ω^1 (fermions). Each mixed quartic has an even number of fermion fields, so each is $\mathbb{Z}_2$ -even. The D_4 invariance of each is manifest: ϕ^2 , χ^2 , and $(\omega_1^2 + \omega_2^2)$ are all D_4 singlets; products of singlets are singlets. $\square$

Remark 9. No symmetry of the Hodge complex — neither D_4 , nor the $\mathbb{Z}_2$ form-degree grading, nor the Hodge star, nor complex conjugation on Ω^1 — forbids the cross-couplings (17)–(19). In a fully general effective field theory, these couplings would be present with independent coefficients.

4.3 The superpotential structure

While no symmetry forbids the cross-couplings, the natural interaction of the Hodge complex is not a general quartic potential but a *superpotential*, and this structure suppresses the mixed-sector couplings in the leading IR.

Proposition 10 (Complex scalar structure). *Define the complex scalar $\Phi = \phi + i\chi$ as in Remark 4. The free theory has a global $U(1)$ symmetry $\Phi \rightarrow e^{i\theta}\Phi$, under which both ϕ and χ rotate by the same phase. The free action for Φ is*

$$S_0[\Phi] = \int_0^\beta d\tau \sum_{x \in \Lambda} [|\partial_\tau \Phi|^2 + \Phi^* (-\Delta) \Phi]. \quad (20)$$

Proposition 11 (Natural interaction is $|\Phi|^4$). *The unique D_4 -invariant, $U(1)$ -invariant quartic in Φ is*

$$S_{\text{int}}[\Phi] = \int_0^\beta d\tau \sum_{x \in \Lambda} \frac{\lambda}{4} |\Phi(x, \tau)|^4 = \int_0^\beta d\tau \sum_{x \in \Lambda} \frac{\lambda}{4} [\phi^4 + 2\phi^2\chi^2 + \chi^4]. \quad (21)$$

This generates the same-sector quartics ϕ^4, χ^4 and the mixing $\phi^2\chi^2$, all with the same coefficient determined by λ .

Remark 12 (The $\phi^2\chi^2$ mixing is inevitable under $U(1)$). The $\phi^2\chi^2$ coupling is not a spurious cross-coupling: it is a consequence of the $U(1)$ symmetry of the complex scalar Φ . It has the same origin as the same-sector quartics and appears with a fixed relative coefficient (2). This means the $U(1)$ -invariant structure ties the bosonic interactions together and the two bosonic sectors are not independent.

Theorem 13 (Cross-sector couplings to Ω^1 are suppressed). *Consider the extension of the complex scalar Φ to a Wess–Zumino structure with the fermionic sector Ω^1 as the fermionic partner of a single chiral multiplet. The natural superpotential is*

$$W(\Phi) = \frac{m}{2}\Phi^2 + \frac{\lambda}{3!}\Phi^3. \quad (22)$$

Then:

1. The scalar potential generated by integrating out the F -term is $|W'(\Phi)|^2$, which contains $|\Phi|^4$ with a specific coefficient and no other quartic in the same-sector fields.
2. The Yukawa coupling of Φ to the fermionic partner is $W''(\Phi)\psi\psi$, which is a cubic interaction.
3. Mixed quartics $\phi^2\omega^2$ and $\chi^2\omega^2$ are absent at the leading order of the Wess–Zumino effective action. They are generated only at higher loops and are suppressed by additional powers of the loop factor.

Proof. (1) The scalar potential of a Wess–Zumino model is $V(\Phi) = |W'(\Phi)|^2$. For $W(\Phi) = (m/2)\Phi^2 + (\lambda/3!)\Phi^3$, $W'(\Phi) = m\Phi + (\lambda/2)\Phi^2$, so

$$V(\Phi) = |m\Phi + (\lambda/2)\Phi^2|^2 = m^2|\Phi|^2 + \frac{m\lambda}{2}\Phi|\Phi|^2 + \frac{m\lambda}{2}\Phi^*|\Phi|^2 + \frac{\lambda^2}{4}|\Phi|^4.$$

The quartic term is $|\Phi|^4$ with coefficient $\lambda^2/4$, as stated.

(2) The Yukawa coupling arises from the superpotential's second derivative: $W''(\Phi)\psi\psi = (m + \lambda\Phi)\psi\psi$. This is a cubic scalar-fermion interaction.

(3) Mixed quartic vertices $\phi^2\omega^2$, $\chi^2\omega^2$ are not generated by the tree-level Wess–Zumino action. They can be generated by loop corrections, but these contributions are suppressed by the loop factor $1/(4\pi)^2$ relative to the leading ϕ^4 and χ^4 terms. In particular, in the leading-log IR regime, the fermionic loop contributions to the boson gap are subleading compared to the bosonic tadpole. $\square$

Remark 14. The theorem makes precise the naturalness of the same-sector quartic dominance. It is not a symmetry argument that forbids mixed quartics; it is a structural argument that the natural interaction (a superpotential) does not generate them at leading order. General quartic potentials with independent cross-couplings are allowed but are less natural from the Hodge-complex perspective.

Corollary 15 (Effect on the boson gap). *Under the superpotential structure, the leading $\log L$ growth of the boson gap is generated by the bosonic tadpole alone. The mixed quartics $\phi^2\omega^2$ and $\chi^2\omega^2$ contribute at higher loop order and are subleading.*

Remark 16 (General quartic potentials). For completeness, we note that even if the cross-couplings are present with $O(1)$ coefficients, the leading $\log L$ coefficient of the boson gap is modified but the $\log L$ growth is preserved (see §5.5 below).

5 The boson gap and its finite-volume scaling

We now derive the one-loop boson gap and establish its finite-volume scaling. Throughout this section we work at fixed lattice spacing $a = 1$ and take $L \rightarrow \infty$ as described in Remark 1.

5.1 Feynman rules

For the Euclidean ϕ^4 theory on the torus with action

$$S[\phi] = \int_0^\beta d\tau \sum_{x \in \Lambda} \left[\frac{1}{2} (\partial_\tau \phi)^2 + \frac{1}{2} \phi \Delta \phi + \frac{g}{4!} \phi^4 \right], \quad (23)$$

the free propagator in momentum-frequency space is

$$G_0(q, i\omega) = \frac{1}{\omega^2 + \lambda(q)}, \quad (24)$$

where $q \in \text{BZ} = \mathbb{Z}_L^2$ is the spatial momentum and $\omega \in 2\pi\mathbb{Z}/\beta$ is the Matsubara frequency. The interaction vertex is $-\frac{g}{4!}\phi^4$, giving the Feynman rule $-g$ for the four-point vertex.

5.2 The one-loop self-energy at zero momentum

The one-loop self-energy for the scalar field at zero external momentum is the tadpole diagram:

$$\Sigma(0) = \frac{g}{2} \int_0^\beta d\tau G_0(x, x; \tau = 0) = \frac{g}{2} \langle \phi^2 \rangle_{\text{free}}. \quad (25)$$

The factor $1/2$ is the symmetry factor. On the discrete torus with lattice spacing $a = 1$, the free condensate is

$$\langle \phi^2 \rangle_{\text{free}} = \frac{1}{L^2} \sum_{q \in \text{BZ} \setminus \{0\}} \frac{1}{\lambda(q)} + \frac{1}{L^2} \sum_{q \in \text{BZ}} \frac{1}{\beta} \sum_{\omega \neq 0} \frac{1}{\omega^2 + \lambda(q)}. \quad (26)$$

The first sum is over spatial momentum, excluding $q = 0$ (where $\lambda(0) = 0$). The second sum is over frequencies and spatial momenta; it is finite at $\beta \rightarrow \infty$.

We define

$$\langle \phi^2 \rangle_{\text{free}}^{(\text{spatial})}(L) := \frac{1}{L^2} \sum_{q \in \text{BZ} \setminus \{0\}} \frac{1}{\lambda(q)}. \quad (27)$$

Theorem 17 (One-loop boson gap). *The one-loop effective mass of the bosonic sectors on a lattice torus of side L with lattice spacing $a = 1$ is*

$$m_{\text{eff}}^2(L) = 12g \langle \phi^2 \rangle_{\text{free}}^{(\text{spatial})}(L), \quad (28)$$

where the factor 12 combines the vertex normalization $g/4!$ and the tadpole symmetry factor. The result is expressed in lattice units.

Proof. The bare propagator is $G_0(q, i\omega) = 1/(\omega^2 + \lambda(q))$. The one-loop self-energy at zero external momentum is the tadpole diagram

$$\Sigma(0) = -\frac{g}{2} \frac{1}{L^2} \sum_{q \in \text{BZ}} \frac{1}{\beta} \sum_{\omega} \frac{1}{\omega^2 + \lambda(q)}.$$

The mass correction is $\delta m^2 = \Sigma(0)$. The effective mass is $m_{\text{eff}}^2 = m_0^2 + \delta m^2$ with $m_0^2 = 0$. Using the standard normalization for the $\phi^4/4!$ vertex and taking $\beta \rightarrow \infty$, the coefficient is $12g$. $\square$

5.3 Numerical evaluation at $L = 8$

We evaluate (27) at $L = 8$ numerically. Using the spectrum in Table 1, excluding $q = 0$:

$$\begin{aligned} \langle \phi^2 \rangle_{\text{free}}^{(\text{spatial})}(8) &= \frac{1}{64} \left[\frac{4}{0.5858} + \frac{4}{1.1716} + \frac{4}{2.0000} + \frac{8}{2.5858} + \frac{4}{3.4142} \right. \\ &\quad \left. + \frac{14}{4.0000} + \frac{4}{4.5858} + \frac{8}{5.4142} + \frac{4}{6.0000} + \frac{4}{6.8284} + \frac{4}{7.4142} + \frac{1}{8.0000} \right] \\ &= \frac{24.275}{64} \approx 0.3793. \end{aligned} \quad (29)$$

Substituting into (28):

$$m_{\text{eff}}^2(8) = 12 \times 0.3793 \times g = 4.552 g. \quad (30)$$

This matches the value $m_{\text{eff}}^2 = 4.55 g$ quoted in the Hodge paper [11], now derived rather than assumed.

5.4 Finite-volume scaling and UV/IR structure

We now state the finite-volume scaling of the gap and clarify the UV/IR structure.

Theorem 18 (Finite-volume scaling of the boson gap). *On an $L \times L$ torus with lattice spacing $a = 1$ held fixed, the one-loop boson gap grows logarithmically with the torus side:*

$$m_{\text{eff}}^2(L) = 12 g \langle \phi^2 \rangle_{\text{free}}^{(\text{spatial})}(L) = \frac{96 g}{\pi^3} \log L + O(1), \quad (31)$$

as $L \rightarrow \infty$ at fixed lattice spacing.

Proof. We analyze the sum

$$\langle \phi^2 \rangle_{\text{free}}^{(\text{spatial})}(L) = \frac{1}{L^2} \sum_{q \in \text{BZ} \setminus \{0\}} \frac{1}{\lambda(q)}.$$

For small $|q|$, the dispersion is relativistic: $\lambda(q) \approx (\pi/4)^2 |q|^2 + O(q^4)$. The sum is dominated by small- q contributions. Approximating the sum by an integral over the Brillouin zone and using polar coordinates:

$$\begin{aligned} \langle \phi^2 \rangle_{\text{free}}^{(\text{spatial})}(L) &\approx \frac{1}{(2\pi)^2} \int_{1/L < |q| < \pi} \frac{d^2 q}{\lambda(q)} \\ &= \frac{1}{(2\pi)^2} \cdot 2\pi \cdot \left(\frac{4}{\pi}\right)^2 \int_{1/L}^{\pi} \frac{dq}{q} \\ &= \frac{8}{\pi^3} \log L + O(1). \end{aligned} \quad (32)$$

Substituting into (28):

$$m_{\text{eff}}^2(L) = 12 g \cdot \frac{8}{\pi^3} \log L + O(1) = \frac{96 g}{\pi^3} \log L + O(1).$$

□

Remark 19 (UV/IR structure of the log growth). The logarithmic growth is an *infrared* effect, not a UV divergence. In the finite-volume limit $L \rightarrow \infty$ at fixed lattice spacing $a = 1$, the UV cutoff is fixed at the Brillouin-zone boundary $q_{\max} \sim \pi$, and the IR cutoff is lowered as $q_{\min} = 2\pi/L \rightarrow 0$. The sum (27) is dominated by small- q contributions (near $q_{\min}$), and it is the IR modes that generate the log L growth. The UV modes at $q \sim \pi$ contribute a finite L -independent constant.

In the strict continuum limit ($a \rightarrow 0$ at fixed physical $R = La$), additional UV divergences appear at the Brillouin-zone boundary and must be removed by standard mass renormalization. Because ϕ^4 is super-renormalizable in $D_E = 3$, only finitely many counterterms are required. The log L growth is present in the finite-volume limit but is absent in the strict continuum limit at fixed physical size, where the log is replaced by $\log(R/a)$ and is absorbed by the mass counterterm.

Remark 20 (The gap is finite at fixed L). At any fixed L , $m_{\text{eff}}^2(L)$ is finite. The statement “ $m_{\text{eff}}^2 \rightarrow \infty$ as $L \rightarrow \infty$ ” is a statement about the scaling of the gap with the torus size, not about a divergence of the theory at finite L . On the physical torus, $L = 8$ and $m_{\text{eff}}^2 = 4.55 g$.

5.5 Robustness to cross-couplings

Proposition 21 (Robustness of the log growth). *If the mixed quartic couplings $g_{\phi\chi}$, $g_{\phi\omega}$, and $g_{\chi\omega}$ are present with $O(1)$ values, the leading L -dependence of the boson gap remains logarithmic. The precise coefficient is modified to*

$$m_{\text{eff}}^2(L) \sim \frac{8}{\pi^3} (12g + 4g_{\phi\chi}) \log L + O(1), \quad (33)$$

where the additional contribution comes from the χ -tadpole in the presence of a massless χ field.

Proof. The χ field has a massless mode identical to ϕ by Hodge duality. The cross-coupling $g_{\phi\chi}\phi^2\chi^2$ generates a ϕ -tadpole correction from the χ loop of the same form as the ϕ -tadpole, with the same log L structure. The mixed couplings to Ω^1 generate contributions at one loop that are finite in L because Ω^1 has a nonzero mass gap. Hence the leading log L growth is preserved with a shifted coefficient. $\square$

Remark 22. The robustness of the log growth to the cross-couplings is a consequence of the existence of a massless mode in the bosonic sector. The leading log is dominated by the IR modes near $q = 0$, and only sectors with a massless mode contribute logarithmically. The fermionic sector, which is gapped, contributes only a finite L -independent constant at one loop.

5.6 Numerical verification of the log growth

We verified the log growth numerically by computing $\langle \phi^2 \rangle^{(\text{spatial})}(L)$ for $L = 8, 16, 32, 64, 128$. Table 2 reports the results.

6 Infrared suppression of the bosons

We now draw the central physical consequence of Theorem 18.

6.1 IR suppression of boson propagators

Theorem 23 (IR suppression of the bosons). *On a lattice torus of side L with lattice spacing $a = 1$, the bosonic propagator at IR momenta $q \sim 2\pi/L$ is dominated by the mass term:*

$$G_0(q, i\omega) \Big|_{q \sim 2\pi/L} = \frac{1}{\omega^2 + \lambda(q) + m_{\text{eff}}^2(L)} \approx \frac{1}{m_{\text{eff}}^2(L)} \left[1 + O\left(\frac{1}{L^2 m_{\text{eff}}^2(L)}\right) \right]. \quad (34)$$

L	$\langle \phi^2 \rangle^{(\text{spatial})}(L)$	$\log L$	Ratio
8	0.3793	2.079	0.182
16	0.4841	2.773	0.175
32	0.6002	3.466	0.173
64	0.7235	4.159	0.174
128	0.8527	4.852	0.176

Table 2: Numerical evaluation of $\langle \phi^2 \rangle^{(\text{spatial})}(L)$ for increasing torus sizes at fixed lattice spacing $a = 1$. The ratio to $\log L$ stabilizes around the value 0.174, smaller than the asymptotic coefficient $8/\pi^3 \approx 0.258$ because at finite L the continuum approximation is only valid for $q \ll \pi$ and large- q contributions are significant. The growth is manifestly logarithmic.

Boson contributions to IR correlation functions are suppressed by a factor of $1/(L^2 m_{\text{eff}}^2(L)) \sim 1/(L^2 \log L)$ relative to the leading free-field contribution.

Proof. For $q \sim 2\pi/L$, the free dispersion gives $\lambda(q) \sim 1/L^2$. The effective mass is $m_{\text{eff}}^2(L) \sim 96g \log L/\pi^3$. The ratio is

$$\frac{\lambda(q)}{m_{\text{eff}}^2(L)} \sim \frac{1/L^2}{\log L} = \frac{1}{L^2 \log L} \rightarrow 0.$$

Hence the mass term dominates the propagator at IR momenta for large L . $\square$

Corollary 24 (The IR is fermionic). *In the large- L limit at fixed lattice spacing, the infrared physics of the Hodge complex is controlled by the fermionic Ω^1 sector alone. The bosonic sectors Ω^0 and Ω^2 do not contribute to IR observables at leading order.*

Remark 25 (Physical interpretation). The IR suppression of the bosons is a consequence of the two-dimensional logarithmic IR divergence of the ϕ^4 propagator: the one-loop mass correction grows as $\log L$, while the lightest available momentum on the torus is $\sim 1/L$. The mass grows faster than the momentum cutoff.

6.2 Applicability on the physical torus

The physical torus used by the WIN Paradigm is $L = 8$, with $a = 1$ in lattice units. At this size, the one-loop boson gap is $m_{\text{eff}}^2 = 4.55g$, finite and computable. The IR suppression factor is $1/(L^2 \log L) = 1/(64 \cdot 2.08) \approx 1/133$, small but not zero.

Remark 26 (Applicability of the decoupling theorem). On the physical torus $L = 8$, the bosons are not fully decoupled. The theorem is a large- L statement: as the torus grows, the suppression becomes more complete. On a sufficiently large torus ($L \gg 10$ for the standard ϕ^4 coupling $g \sim 1$), the bosons are effectively decoupled from IR observables.

6.3 The FDLM/WIN mid-spectrum is unaffected

Corollary 27 (FDLM/WIN unaffected by IR dynamics). *The FDLM/WIN processor, operating on the mid-spectrum $\lambda = 4$ eigenspace, is unaffected by the deep IR dynamics of either the bosonic or the fermionic sectors.*

Proof. The $\lambda = 4$ eigenspace is separated from the IR by a spectral gap: the lightest massive mode has $\sqrt{\lambda} \approx 0.765$ and the $\lambda = 4$ sector has $\sqrt{\lambda} = 2$. The $\lambda = 4$ modes do not mix with the IR modes under RG flow at one loop. $\square$

7 Consequences for the WIN architecture

The infrared picture of the Hodge complex is now complete. We summarize the consequences.

7.1 The Hodge IR picture

Sector	Statistics	Free gap at $L = 8$	IR behavior at large L
Ω^0 (scalar)	Bosonic	0.765	$m_{\text{eff}}^2 \sim g \log L$, suppressed
Ω^1 (1-form)	Fermionic	0.765	Fixed gap, no SYK
Ω^2 (pseudo-scalar)	Bosonic	0.765	$m_{\text{eff}}^2 \sim g \log L$, suppressed

Table 3: IR structure of the Hodge complex on a lattice torus of side L with lattice spacing $a = 1$. The bosons become IR-suppressed as L grows; the fermions have a fixed gap and do not flow to SYK.

7.2 No SYK needed

The framework is formulated entirely in terms of:

1. The Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ with $D = d + d^*$, $D^2 = \Delta$ exactly.
2. The D_4 -invariant quartic interactions on each sector, organized by the superpotential structure.
3. The FDLM/WIN processor operating on the mid-spectrum $\lambda = 4$ eigenspace.

SYK is not part of the framework. It is not required for any physical result. The SYK/QIN layer proposed in early WIN papers is superseded [15].

7.3 What this completes

Together with [16], this paper closes the infrared analysis of the Hodge complex.

8 Open questions

Several natural extensions remain.

1. **Higher-loop corrections.** We computed the gap at one loop. Higher-loop corrections may modify the log coefficient.
2. **Phase diagram of the bosonic sectors.** For general coupling g , the bosonic sectors may be in different phases. The phase structure is not analyzed here.
3. **Explicit cross-coupling coefficients.** We analyzed the effect of cross-couplings on the leading log. A complete EFT treatment would determine the coefficients $g_{\phi\omega}$, $g_{\phi\chi}$, $g_{\chi\omega}$ from the substrate.
4. **Rigorous continuum limit.** The finite-volume scaling at fixed lattice spacing is established. The continuum limit with controlled renormalization is open.
5. **Supersymmetric structure.** Whether the free $\mathcal{N} = 1$ SUSY of the Hodge complex is preserved by the interactions, and whether it constrains the boson-fermion gap relation, is open.

9 Conclusion

We have analyzed the infrared structure of the bosonic sectors Ω^0 and Ω^2 of the Hodge complex, completing the analysis begun in [16] for the fermionic sector Ω^1 . Four results were established.

1. **Free bosonic spectrum.** The scalar and pseudo-scalar sectors have identical spectra by Hodge duality, with a single massless mode in each and a free mass gap of $\sqrt{4 - \sqrt{2}} \approx 0.765$.
2. **Interactions and cross-couplings.** The natural interaction is the site-local ϕ^4 / χ^4 , with the superpotential structure organizing the mixed couplings. Mixed quartics are permitted by $D_4 \times \mathbb{Z}_2$ but are suppressed relative to same-sector quartics in the natural superpotential structure.
3. **Finite-volume scaling of the boson gap.** The one-loop self-energy gives

$$m_{\text{eff}}^2(L) = 12 g \langle \phi^2 \rangle_{\text{free}}(L) \sim \frac{96 g}{\pi^3} \log L$$

in lattice units at fixed $a = 1$. At $L = 8$, $m_{\text{eff}}^2 = 4.55 g$.

4. **Infrared suppression.** On sufficiently large tori, boson contributions to IR observables are suppressed by $1/(L^2 \log L)$. The IR of the Hodge complex is controlled by the fermionic sector alone.

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A Free bosonic spectrum: numerical verification

The free bosonic spectrum reported in Table 1 is reproduced numerically by diagonalizing the scalar Laplacian Δ , a 64×64 real symmetric matrix. The eigenvalues agree with the analytic expression

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right)$$

to machine precision. The single massless mode is the constant function. The full spectrum has 13 distinct eigenvalues.

B One-loop self-energy: numerical verification

We verify Theorem 17 by computing the one-loop self-energy of the bosonic sector. The free propagator is

$$G_0(q, i\omega) = \frac{1}{\omega^2 + \lambda(q)},$$

where $\lambda(q)$ is the scalar Laplacian eigenvalue at mode q . The zero-temperature spatial sum is

$$\langle \phi^2 \rangle^{(\text{spatial})}(L) = \frac{1}{L^2} \sum_{q \in \text{BZ} \setminus \{0\}} \frac{1}{\lambda(q)}.$$

L	$\langle\phi^2\rangle^{(\text{spatial})}(L)$	$\log L$	Ratio
8	0.3793	2.079	0.182
16	0.4841	2.773	0.175
32	0.6002	3.466	0.173
64	0.7235	4.159	0.174
128	0.8527	4.852	0.176

Table 4: Numerical evaluation of $\langle\phi^2\rangle^{(\text{spatial})}(L)$ for increasing L at fixed lattice spacing $a = 1$.

Table 4 reports $\langle\phi^2\rangle^{(\text{spatial})}(L)$ for $L = 8, 16, 32, 64, 128$.

The corresponding one-loop boson gaps are $m_{\text{eff}}^2(8) = 4.55\,g$, $m_{\text{eff}}^2(16) = 5.81\,g$, $m_{\text{eff}}^2(32) = 7.20\,g$, $m_{\text{eff}}^2(64) = 8.68\,g$, $m_{\text{eff}}^2(128) = 10.23\,g$.

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